CSIAM 2026

On the Scaling of Neural Optimizers

A L-Smoothness Perspective

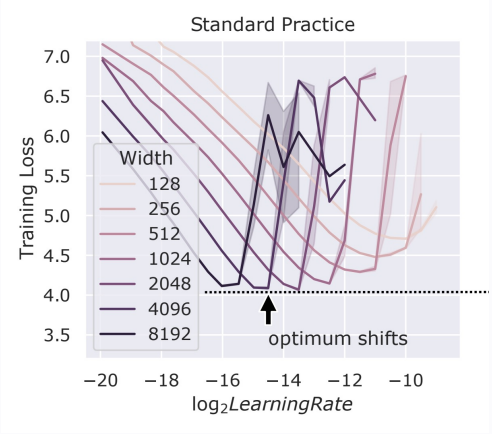
Yiping Lu

Beijing International Center for Mathematical Research
Peking University

Joint work with Ruihan Xu (University of Chicago) and Jiajin Li (UBC)

$\|\cdot\|_{p\to q}$

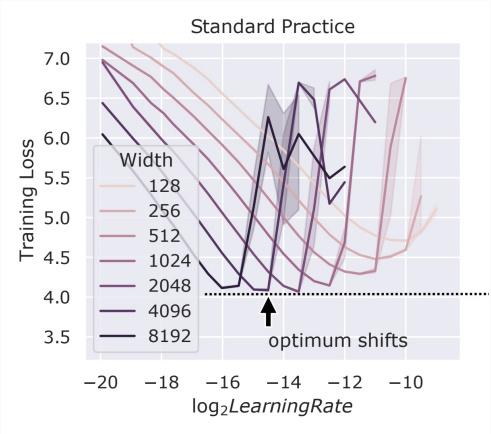
Hyperparameter Transfer [Yang et al., NeurIPS 2021]



Key empirical observations

- The optimal learning rate **decreases as network width increases**.
- A learning rate tuned at width **512** can **diverge** or **slow dramatically** when scaled to width **2048**.

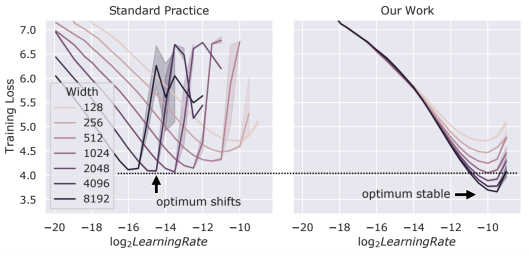
Hyperparameter Transfer [Yang et al., NeurIPS 2021]



Why this matters for LLMs

Pretraining state of the art base models can cost **hundreds of millions of dollars**. Hyperparameter transfer is therefore **essential**, not optional.

Hyperparameter Transfer [Yang et al., NeurIPS 2021]



Yang et al., NeurIPS 2021

`Adam` learning rate scaling:

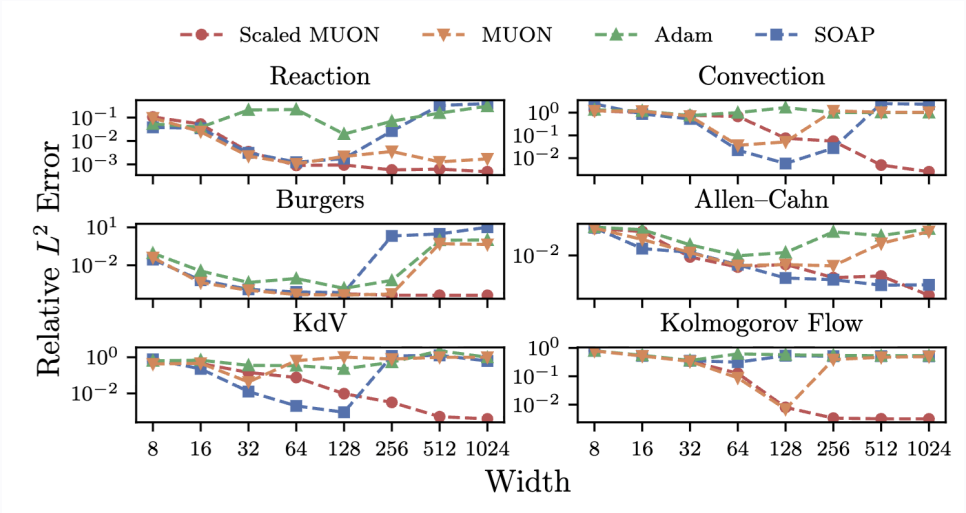
- **Weight matrices**
  - Scale learning rate by $\frac{1}{\texttt{d_in}}$
- **Embedding matrices**
  - No learning-rate scaling

Here, `d_in` denotes the effective input dimensionality of the parameter block.

Current theory is highly specialized

Existing derivations require extensive random matrix arguments and justify transfer only near initialization.

Large PINN is Hard to Train



What this talk is about

Design **neural optimizers that scale in width and depth** *and* develop a principled understanding of **learning-rate scaling rules** from a **matrix operator norm perspective**.

Problem Setup: Neural Network Optimization

We consider the constrained optimization problem

$$\min_{\boldsymbol{W}_{1:\ell},\, \boldsymbol{b}_{1:\ell} \in \mathcal{C}} \; f(\boldsymbol{W}_{1:\ell}, \boldsymbol{b}_{1:\ell}) \;:=\; \mathcal{L}\big(\boldsymbol{y}_\ell(\boldsymbol{x})\big),$$

where $\boldsymbol{y}_\ell(\boldsymbol{x}) \in \mathbb{R}$ is the output of an ℓ-layer feedforward neural network evaluated at input $\boldsymbol{x} \in \mathbb{R}^d$.

Network architecture

The network is defined recursively as

$$\boldsymbol{y}_i(\boldsymbol{x}) \;:=\; \sigma\big(\boldsymbol{W}_i\, \boldsymbol{y}_{i-1}(\boldsymbol{x}) + \boldsymbol{b}_i\big), \qquad i = 1, \ldots, \ell,$$

with

$$\boldsymbol{W}_1 \in \mathbb{R}^{w \times d} \quad \text{and} \quad \boldsymbol{W}_i \in \mathbb{R}^{w \times w} \;\; (i \geq 2).$$

Later, we write $\Theta := \{\boldsymbol{W}_{1:\ell},\, \boldsymbol{b}_{1:\ell}\}$ for simplicity.

Constrained Optimization and Frank Wolfe Updates

Constrained training problem

$$\min_{\Theta \in \mathcal{C}_R} f(\Theta), \quad \mathcal{C}_R = \{\Theta : \|\Theta\| \leq R\}, \quad R = 1/\lambda > 0.$$

Frank Wolfe updates with decoupled weight decay

For a gradient or momentum estimate M^t, set $\alpha_t = \eta_t \lambda$ and compute

$$S^t \in \arg\min_{S \in \mathcal{C}_R} \langle M^t, S \rangle, \qquad d^t = -S^t / R.$$

$$\Theta^{t+1} = (1 - \alpha_t)\Theta^t + \alpha_t S^t = \Theta^t - \eta_t d^t - \underbrace{\eta_t \lambda \Theta^t}_{\text{decoupled weight decay}}.$$

Lemma (Uniform bound along the entire trajectory)

If $\|\Theta^0\| \leq R$, $\|d^t\| \leq \lambda R$, *and* $0 \leq \eta_t \lambda \leq 1$ *for every* $t \geq 0$, *then* $\|\Theta^t\| \leq R$ *for all* $t \geq 0$.

Exact gradients give standard Frank Wolfe. Momentum gives a momentum variant.

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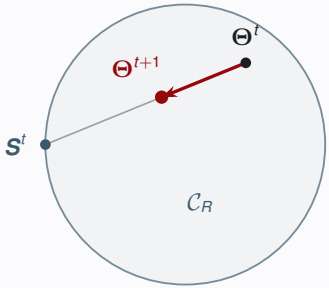
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Exact gradients give standard Frank Wolfe. Momentum gives a momentum variant.

Decoupled Weight Decay as a Frank Wolfe Step

$$\min_{\Theta \in \mathcal{C}_R} f(\Theta), \qquad \mathcal{C}_R = \{\Theta : \|\Theta\| \le R\}, \qquad R = 1/\lambda.$$



A convex combination stays feasible.

Choose a feasible linear minimizer

$$S^t \in \arg\min_{S \in \mathcal{C}_R} \langle \nabla f(\Theta^t), S \rangle, \qquad d^t = -S^t/R.$$

Move partway toward it

$$\Theta^{t+1} = (1 - \alpha_t)\Theta^t + \alpha_t S^t.$$

With $\alpha_t = \eta_t \lambda \in [0, 1]$,

$$\Theta^{t+1} = \Theta^t - \eta_t d^t - \underbrace{\eta_t \lambda \Theta^t}_{\text{decoupled weight decay}}.$$

Weight decay supplies the contraction in a Frank Wolfe update.

Exact for the norm ball oracle and a feasible start. Momentum gives a variant. Generic AdamW need not use this oracle.

AdamW and signSGD

- AdamW is a standard optimizer for large scale neural network training. [Kingma and Ba, 2015; Loshchilov and Hutter, 2019]

$$
\begin{aligned}
\boldsymbol{g}^t &= \nabla f(\boldsymbol{\theta}^{t-1}, \xi_t) \quad \text{(gradient)} \\
\boldsymbol{m}^t &= \beta_1 \boldsymbol{m}^{t-1} + (1-\beta_1)\boldsymbol{g}^t \quad \text{(first moment)} \\
\boldsymbol{v}^t &= \beta_2 \boldsymbol{v}^{t-1} + (1-\beta_2)\boldsymbol{g}^t \odot \boldsymbol{g}^t \quad \text{(second moment)} \\
\hat{\boldsymbol{m}}^t &= \frac{\boldsymbol{m}^t}{1-\beta_1^t}, \qquad \hat{\boldsymbol{v}}^t = \frac{\boldsymbol{v}^t}{1-\beta_2^t} \\
\boldsymbol{\theta}^t &= \boldsymbol{\theta}^{t-1} - \eta \left(\frac{\hat{\boldsymbol{m}}^t}{\sqrt{\hat{\boldsymbol{v}}^t} + \epsilon} \right) - \eta\lambda\boldsymbol{\theta}^{t-1}
\end{aligned}
$$

Adam approximates `signSGD` [Bernstein et al, 2018, ICML]. $\boldsymbol{\theta}^t \approx \boldsymbol{\theta}^{t-1} - \eta \frac{\boldsymbol{g}^t}{\sqrt{\boldsymbol{g}^t \odot \boldsymbol{g}^t}} \approx \boldsymbol{\theta}^{t-1} - \eta \cdot \operatorname{sign}(\boldsymbol{g}^t)$.

Lion: Sign Updates with Momentum [Chen et al., 2023]

Key idea

Lion can be viewed as **SignSGD + momentum + decoupled weight decay**.

$$
\begin{aligned}
\boldsymbol{g}^t &= \nabla f(\boldsymbol{\theta}^{t-1}, \xi_t) \\
\boldsymbol{m}^t &= \beta_1 \boldsymbol{m}^{t-1} + (1-\beta_1)\boldsymbol{g}^t \\
\boldsymbol{\theta}^t &= \boldsymbol{\theta}^{t-1} - \eta \operatorname{sign}(\boldsymbol{m}^t) - \eta\lambda\boldsymbol{\theta}^{t-1}
\end{aligned}
$$

- Automatic discovery of optimizers through RL and meta learning
- Lower memory usage than AdamW

Muon: Matrix Sign Updates

Key idea

Change the optimization geometry from a **vector space** to a **matrix space**.

$$
\begin{aligned}
\boldsymbol{G}^t &= \nabla f(\boldsymbol{\Theta}^{t-1}, \xi_t) \\
\boldsymbol{M}^t &= \beta_1 \boldsymbol{M}^{t-1} + (1-\beta_1)\boldsymbol{G}^t \\
\boldsymbol{\Theta}^t &= \boldsymbol{\Theta}^{t-1} - \eta \,\mathrm{matrixsign}(\boldsymbol{M}^t) - \eta\lambda\boldsymbol{\Theta}^{t-1}
\end{aligned}
$$

- If $\boldsymbol{G} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$, then $\mathrm{matrixsign}(\boldsymbol{G}) = \boldsymbol{U}\boldsymbol{V}^\top$; it can be approximated by **Newton Schulz iterations**. [Jordan et al., 2024]

Competing Learning Rate Scaling Rules for Muon

μP variant [Yang et al., 2023]

$$\sqrt{\frac{\texttt{d_out}}{\texttt{d_in}}} \cdot \underbrace{\eta_{\texttt{base}}}_{\text{optimal LR for the base (small) model}}$$

Grafting in spectral geometry.

Moonlight variant [Liu et al., 2025]

$$\sqrt{\max\{\texttt{d_out}, \texttt{d_in}\}} \cdot \underbrace{\eta_{\texttt{Adam}}}_{\text{tuned LR using AdamW}}$$

- Deployed in Kimi
- Euclidean grafting [Agarwal et al., 2020]
- Optimal learning rate decays as $O(1/\sqrt{\text{width}})$.

Here, `d_out` denotes the effective output dimensionality of the parameter block.

Steepest Descent in Matrix Operator Geometry

Matrix Thinking

All these optimizers admit a unified interpretation as steepest descent under different matrix operator norms.

$$\boxed{\boldsymbol{D}^t = \underset{\|\boldsymbol{D}\|_? \le 1}{\arg\min} \langle \boldsymbol{M}^t, \boldsymbol{D} \rangle} \qquad (\star)$$

Definition (Operator Norm)

Let $\|\cdot\|_{\text{in}}$ and $\|\cdot\|_{\text{out}}$ be norms on $\mathbb{R}^n$ and $\mathbb{R}^m$. The operator norm of $\boldsymbol{D}$ induced by these norms is defined as

$$\|\boldsymbol{D}\|_{\text{in}\to\text{out}} := \sup_{\|\boldsymbol{x}\|_{\text{in}}=1} \|\boldsymbol{D}\boldsymbol{x}\|_{\text{out}}.$$

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Choosing the Matrix Operator Norm

Example (SignSGD/Adam/Lion, $\ell_1 \to \ell_\infty$)

The element-wise ℓ_∞ norm of $\boldsymbol{D}$ coincides with its $\ell_1 \to \ell_\infty$ operator norm, i.e.,

$$\|\boldsymbol{D}\|_{1\to\infty} = \max_{i,j} |\boldsymbol{D}_{i,j}| \quad \text{and} \quad \boldsymbol{D}^* = \operatorname*{arg\,min}_{\|\boldsymbol{D}\|_{1\to\infty}\le 1} \langle \boldsymbol{M}, \boldsymbol{D}\rangle = -\mathtt{sign}(\boldsymbol{M}).$$

Example (Muon, $\ell_2 \to \ell_2$)

The spectral norm of $\boldsymbol{D}$ coincides with its $\ell_2 \to \ell_2$ operator norm, i.e.,

$$\|\boldsymbol{D}\|_{2\to 2} = \|\boldsymbol{D}\|_2 \quad \text{and} \quad \boldsymbol{D}^* = \operatorname*{arg\,min}_{\|\boldsymbol{D}\|_{2\to 2}\le 1} \langle \boldsymbol{M}, \boldsymbol{D}\rangle = -\mathtt{matrixsign}(\boldsymbol{M}).$$

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Column Normalization [Bernstein and Newhouse, 2024]

Example (Geometry: $\ell_1 \to \ell_q$)

$$\|\boldsymbol{D}\|_{1\to q} = \max_{c\in[n]} \|\boldsymbol{D}_{:,c}\|_q,$$

$$\boldsymbol{D}^\star = -\mathrm{colnorm}_q(\boldsymbol{M}).$$

$$\mathrm{colnorm}_q(\boldsymbol{M})_{:,c} := \frac{\mathrm{sign}(\boldsymbol{M}_{:,c}) \odot |\boldsymbol{M}_{:,c}|^{q^*-1}}{\|\boldsymbol{M}_{:,c}\|_{q^*}^{q^*-1}},$$

where $\frac{1}{q} + \frac{1}{q^*} = 1$.

Interpretation

Each column is normalized in the dual geometry induced by ℓ_q.

Row Normalization [Bernstein and Newhouse, 2024]

Example (Geometry: $\ell_p \to \ell_\infty$)

$$\|\boldsymbol{D}\|_{p\to\infty} = \max_{r\in[m]} \|\boldsymbol{D}_{r,:}\|_{p^*},$$

$$\boldsymbol{D}^\star = -\mathrm{rownorm}_p(\boldsymbol{M}).$$

$$\mathrm{rownorm}_p(\boldsymbol{M})_{r,:} := \frac{\mathrm{sign}(\boldsymbol{M}_{r,:}) \odot |\boldsymbol{M}_{r,:}|^{p-1}}{\|\boldsymbol{M}_{r,:}\|_p^{p-1}},$$

where $\frac{1}{p} + \frac{1}{p^*} = 1$.

Interpretation

Each row is normalized in the dual geometry induced by ℓ_p.

Matrix Thinking

SignSGD / AdamW	Column norm	Row norm	Muon
$\|\cdot\|_{1\to\infty}$	$\|\cdot\|_{1\to q}$	$\|\cdot\|_{p\to\infty}$	$\|\cdot\|_{2\to 2}$

Choosing the right matrix operator norm is the key to designing new optimizers.

Geometry-Aware Principles for Optimizer Design

- **Low per-iteration computational cost**
  - Choose matrix operator geometries $\|\cdot\|_{p\to q}$ with $p \le q$.
- **Neural network landscape**
  - **Width-independent Lipschitz bound:** the network's *forward sensitivity* is bounded independently of width.
  - **Width-independent smoothness bound:** the *second-order sensitivity* is bounded independently of width.

These width independent bounds naturally induce **learning rate scaling rules** that remain stable across widths and enable hyperparameter transfer.

Why Standard Operator Norms Do Not Compose

Observation

Standard $\ell_p \to \ell_q$ operator norms introduce width dependent factors when they are composed across layers.

For one layer, operator control gives

$$\left\|\nabla_{\boldsymbol{W}_i} \boldsymbol{y}_{\ell+1}\right\|_{\text{out}} \leq \|\boldsymbol{W}_\ell\|_{\text{in}\to\text{out}} \|\nabla_{\boldsymbol{W}_i} \boldsymbol{y}_\ell\|_{\text{in}}.$$

Cross layer requirement

To iterate without a dimensional penalty, require $\|\cdot\|_{\text{in}} \leq \|\cdot\|_{\text{out}}$.

$$\left\|\nabla_{\boldsymbol{W}_i} \boldsymbol{y}_{\ell+2}\right\|_{\text{out}} \leq \prod_{k=\ell}^{\ell+1} \|\boldsymbol{W}_k\|_{\text{in}\to\text{out}} \|\nabla_{\boldsymbol{W}_i} \boldsymbol{y}_\ell\|_{\text{in}}$$

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A Width Dependent Embedding Gap

The two consecutive bounds involve different norms of the same intermediate feature:

$$\left\|\nabla_{\boldsymbol{W}_i} \boldsymbol{y}_{\ell+1}\right\|_{\text{out}} \leq \|\boldsymbol{W}_\ell\|_{\text{in}\to\text{out}} \|\nabla_{\boldsymbol{W}_i} \boldsymbol{y}_\ell\|_{\text{in}},$$
$$\left\|\nabla_{\boldsymbol{W}_i} \boldsymbol{y}_{\ell+2}\right\|_{\text{out}} \leq \|\boldsymbol{W}_{\ell+1}\|_{\text{in}\to\text{out}} \left\|\nabla_{\boldsymbol{W}_i} \boldsymbol{y}_{\ell+1}\right\|_{\text{in}}.$$

Example (The all ones vector)

For $\boldsymbol{x} = (1, \ldots, 1) \in \mathbb{R}^n$,

$$\|\boldsymbol{x}\|_\infty = 1,$$
$$\|\boldsymbol{x}\|_p = n^{1/p},$$
$$\|\boldsymbol{x}\|_1 = n.$$

Consequence

Standard norm embeddings inject powers of the width into the stability bound.

Mean Normalized Norms Compose Across Layers

Definition (Mean-Normalized Norm)

We introduce a width aware geometry through the mean-normalized norms $\|\cdot\|_{(p,\mathtt{mean})}$:

$$\|\boldsymbol{x}\|_{(p,\mathtt{mean})} := \left(\frac{1}{n}\sum_{i=1}^{n}|\boldsymbol{x}_i|^p\right)^{1/p} = n^{-1/p}\|\boldsymbol{x}\|_p.$$

- The factor $n^{-1/p}$ precisely cancels the dimensional scaling of the ℓ_p embeddings.

Fact

For $\boldsymbol{x} \in \mathbb{R}^n$ and $1 \le p \le q \le \infty$,

$$\|\boldsymbol{x}\|_{(p,mean)} \le \|\boldsymbol{x}\|_{(q,mean)}.$$

- Choose $\|\cdot\|_{(p,\mathtt{mean})\to(q,\mathtt{mean})}$ with $p \le q$.

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Width Independent Lipschitz Bound

Theorem (Lipschitz bound in mean normalized geometry)

Assume $1 \le p \le q < \infty$*, bounded inputs, and bounded first derivatives of* σ *and* $\mathcal{L}$*. Define*

$$\Omega_C := \{\Theta : \|\Theta\|_{\text{block}} \le C\},$$

where

$$\|\Theta\|_{\text{block}} := \max\left\{ \|W_1\|_{1\to(q,\text{mean})}, \max_{2\le i\le \ell-1} \|W_i\|_{(p,\text{mean})\to(q,\text{mean})}, \|W_\ell\|_{(p,\text{mean})\to\infty}, \max_i \|b_i\|_\infty \right\}.$$

Then the loss is M Lipschitz on Ω_C*, where M is independent of the width.*

Implication

The forward sensitivity of the network remains controlled as width grows.

Width Independent Smoothness

Definition (L smoothness)

With respect to a norm $\|\cdot\|$, the gradient satisfies

$$\|\nabla f(\boldsymbol{Z}) - \nabla f(\boldsymbol{Z}')\|_* \leq L\|\boldsymbol{Z} - \boldsymbol{Z}'\|.$$

Theorem (Width dependence of the smoothness constant)

Under the preceding boundedness assumptions, with bounded first and second derivatives of σ and $\mathcal{L}$,

$$\|\nabla f(\Theta) - \nabla f(\Theta')\|_{\mathrm{block},*} \leq L\, w^{\max\{0,\, 2/q - 1/p\}} \|\Theta - \Theta'\|_{\mathrm{block}},$$

where L is independent of width.

Width independent smoothness holds when $q \geq 2p$.

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Why the Condition $q \geq 2p$ Appears

Featurewise nonlinearities produce a Hadamard product in the quadratic Taylor term.

Lemma (Mean normalized Hadamard inequality)

For $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ and $p, q \geq 1$,

$$\|\boldsymbol{x} \odot \boldsymbol{y}\|_{(p,\mathrm{mean})} \leq n^{\max\{0,\, 2/q - 1/p\}} \|\boldsymbol{x}\|_{(q,\mathrm{mean})} \|\boldsymbol{y}\|_{(q,\mathrm{mean})}.$$

Consequently,

$$\begin{aligned} \left|\nabla^2 f(\boldsymbol{W})[\Delta \boldsymbol{W}^1, \Delta \boldsymbol{W}^2]\right| &\lesssim \|(\Delta \boldsymbol{W}^1 \boldsymbol{x}) \odot (\Delta \boldsymbol{W}^2 \boldsymbol{x})\|_{(p,\mathrm{mean})} \\ &\lesssim w^{\max\{0,\, 2/q - 1/p\}} \|\Delta \boldsymbol{W}^1 \boldsymbol{x}\|_{(q,\mathrm{mean})} \|\Delta \boldsymbol{W}^2 \boldsymbol{x}\|_{(q,\mathrm{mean})}. \end{aligned}$$

Operator control

If $\|\Delta \boldsymbol{W}^1\|_{(p,\mathrm{mean})\to(q,\mathrm{mean})} \leq 1$, then $\|\Delta \boldsymbol{W}^1 \boldsymbol{x}\|_{(q,\mathrm{mean})} \leq \|\boldsymbol{x}\|_{(p,\mathrm{mean})}$.

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Width Aware Learning Rate Scaling

Matrix Operator Geometry Aware (MOGA) Scaling

For $\boldsymbol{D} \in \mathbb{R}^{\texttt{d_out} \times \texttt{d_in}}$ and $1 \le p \le q \le \infty$,

$$\|\boldsymbol{D}\|_{(p,\texttt{mean})\to(q,\texttt{mean})} = \boxed{\frac{\texttt{d_in}^{1/p}}{\texttt{d_out}^{1/q}}} \, \|\boldsymbol{D}\|_{p\to q}.$$

- At the algebraic endpoint for `Adam` ($\ell_1 \to \ell_\infty$), MOGA recovers the μP factor $\boxed{1/\texttt{d_in}}$.
- For `Muon` ($\ell_2 \to \ell_2$), MOGA recovers the μP factor $\boxed{\sqrt{\texttt{d_out}/\texttt{d_in}}}$.
  - The spectral condition of Yang et al. (2023) imposes the same update scale.
  - When $p = q = 2$, $L = \mathcal{O}(\sqrt{w})$; a stable step size therefore scales as $\mathcal{O}(1/\sqrt{w})$, matching Moonlight.

MOGA Optimizer

Part I Initialization and column geometry

Algorithm 1 MOGA: Matrix-Operator-Geometry-Aware Steepest Descent

Require: Learning rates $\{\eta^t\}_{t\geq 0}$, momentum parameters $\beta_1, \beta_2 \in (0,1)$, initial parameters $\boldsymbol{\Theta}^0$

$\boldsymbol{M}^0 \leftarrow \mathbf{0}$ ▷ $\boldsymbol{M}^t$ has the same block structure as $\boldsymbol{\Theta}^t$

for $t = 1, 2, \ldots$ **do**

  Sample stochastic gradient: $\boldsymbol{G} \leftarrow \nabla F(\boldsymbol{\Theta}^{t-1}; \xi)$, $\xi \sim \mathbb{P}$

  Exponential moving average: $\boldsymbol{M}^t \leftarrow \beta_1 \boldsymbol{M}^{t-1} + (1-\beta_1)\boldsymbol{G}$

  Lookahead (Nesterov-type) momentum: $\tilde{\boldsymbol{M}}^t \leftarrow \beta_2 \boldsymbol{M}^{t-1} + (1-\beta_2)\boldsymbol{G}$

  if Descent under $(1, \text{mean}) \to (q, \text{mean})$ **then** ▷ $q \geq 2$

    for $i = 1$ to ℓ **do**

$$\boldsymbol{W}_i^t \leftarrow \boldsymbol{W}_i^{t-1} - \eta^t \cdot \boxed{\frac{\left(d_{\text{out}}^i\right)^{1/q}}{d_{\text{in}}^i}} \cdot \text{colnorm}_q\left(\tilde{\boldsymbol{M}}^t_{\boldsymbol{W}_i^{t-1}}\right)$$

$$\boldsymbol{b}_i^t \leftarrow \boldsymbol{b}_i^{t-1} - \eta^t \,\text{sign}\left(\tilde{\boldsymbol{M}}^t_{\boldsymbol{b}_i}\right)$$ ▷ or steepest descent under (q, mean) norm

    end for

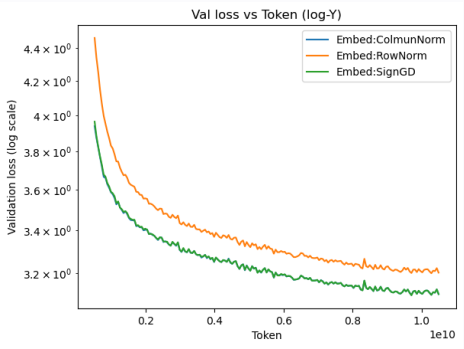
MOGA Optimizer

Part II Row geometry and loop closure

else if Descent under $(p, \text{mean}) \to \infty$ **then**
  for $i = 1$ to ℓ **do**
    $\boldsymbol{W}_i^t \leftarrow \boldsymbol{W}_i^{t-1} - \eta^t \cdot \boxed{\frac{1}{(d_{\text{in}}^i)^{1/p}}} \cdot \text{rownorm}_p\left(\tilde{\boldsymbol{M}}^t_{\boldsymbol{W}_i^{t-1}}\right)$
    $\boldsymbol{b}_i^t \leftarrow \boldsymbol{b}_i^{t-1} - \eta^t \,\text{sign}\left(\tilde{\boldsymbol{M}}^t_{\boldsymbol{b}_i}\right)$
  end for
end if
end for

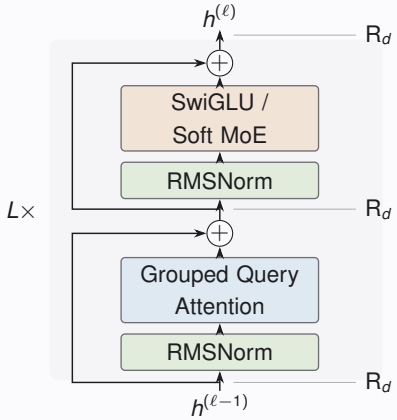
Adapting to Transformers

- **Linear layers in MLPs and attention:** optimized using MOGA with scaling.
- **Biases and LayerNorm parameters:** optimized using SignGD.
- **Input word and positional embeddings:** map one-hot vectors (with ℓ_1 geometry) to feature representations (with (q, mean) geometry).
- **Word unembedding:** the unembedding matrix is tied to the word embedding matrix.



Embedding and output layer comparison

The Residual Stream Stays in RMS Space



Feature spaces

R_n: RMS size C_n: largest coordinate
$A_{H,r}$: largest head RMS (Q/K)

$H_{H,r}$: RMS across all heads (V/O)

The geometry is local to each module

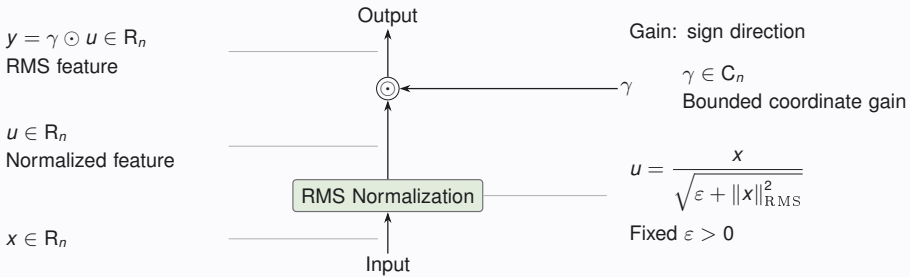
Q/K: Muon per head V/O: whole matrix Muon

Up/Down: Muon Gate/router: RowNorm

Sequence features use the largest token norm.
Fixed depth and bounded parameters; full assumptions in backup.

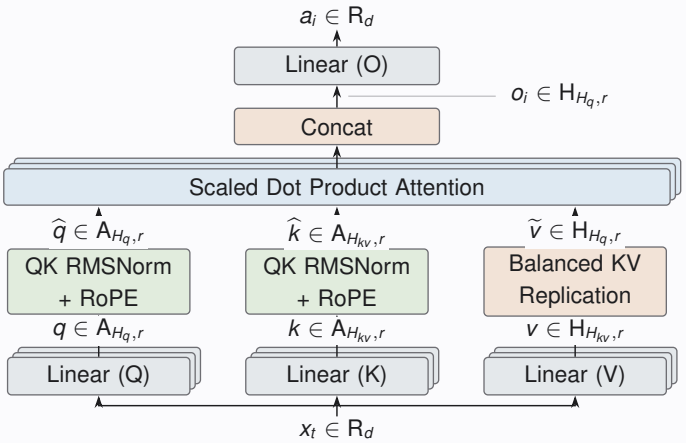
Each branch returns to RMS space. Stronger norms are needed only inside selected modules.

RMSNorm Rescales a Vector by One Shared Number



RMSNorm averages over coordinates. The gain alone needs coordinatewise control.

Q/K Use Headwise Muon; V/O Use Whole Matrix Muon



Whole matrix Muon

$W_V : \mathrm{R}_d \to \mathrm{H}_{H_{kv},r}$

$W_O : \mathrm{H}_{H_q,r} \to \mathrm{R}_d$

Muon within each head

$W_Q : \mathrm{R}_d \to \mathrm{A}_{H_q,r}$

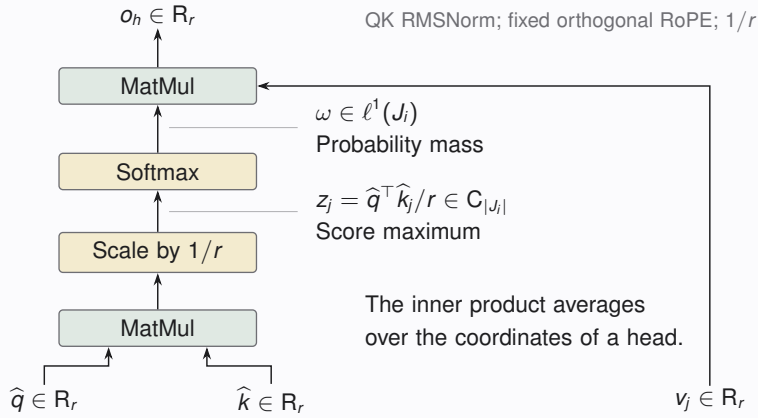
$W_K : \mathrm{R}_d \to \mathrm{A}_{H_{kv},r}$

$\mathrm{A}_{H,r}$: head maximum

$\mathrm{H}_{H,r}$: head RMS

Q/K control each head separately. V/O preserve the RMS of the full feature vector.

Softmax Acts on Scores, Not on Q/K Coordinates

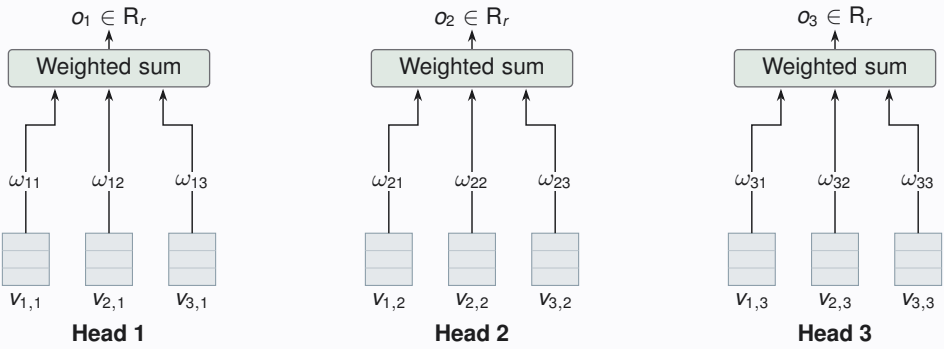


The score map turns head RMS control into the maximum norm that softmax needs.

Bounded Attention Weights Preserve Value RMS

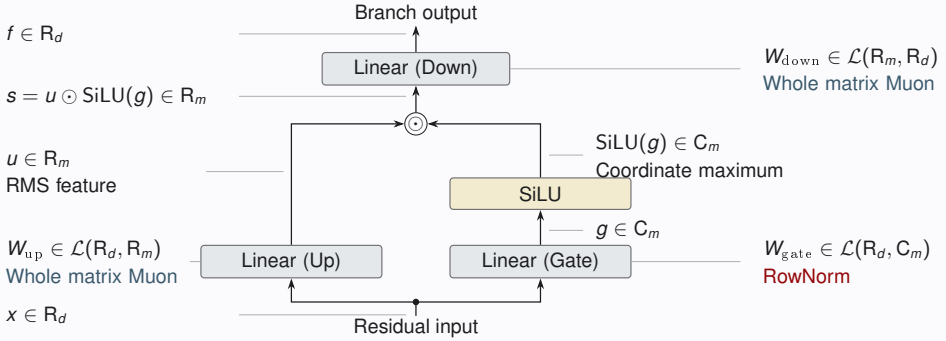
$\omega_{hj} \le D\mu(j)$ one shared key distribution

$o \in \mathrm{H}_{H,r}$ RMS across all heads



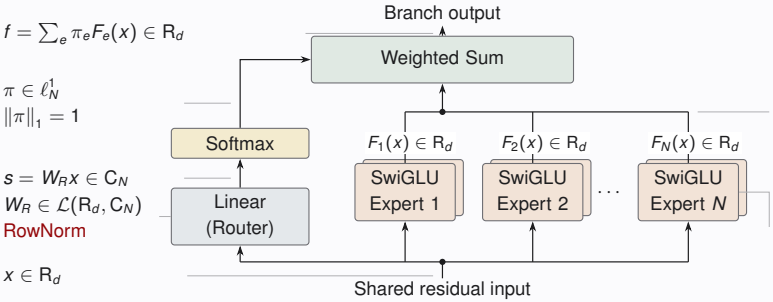
A shared bound on attention weights preserves RMS across heads.

SwiGLU Needs Stronger Control Only on the Gate



The gate multiplies coordinates directly. Its maximum norm protects the RMS of the Up branch.

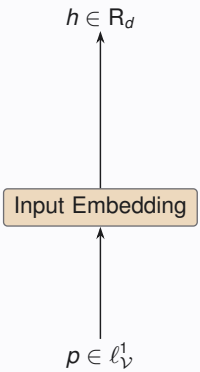
Soft MoE Pairs Probability Mass with Expert RMS

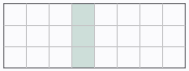


Softmax redistributes a fixed probability mass. The mixture is controlled by the largest expert RMS.

Embedding and Readout Use Different Geometries

Embedding



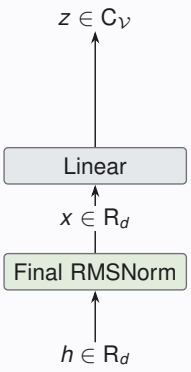


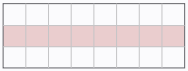
$E : \ell^1_{\mathcal{V}} \to \mathrm{R}_d$

ColumnNorm

One token, one column.

Readout





$W_{\mathrm{lm}} : \mathrm{R}_d \to \mathrm{C}_{\mathcal{V}}$

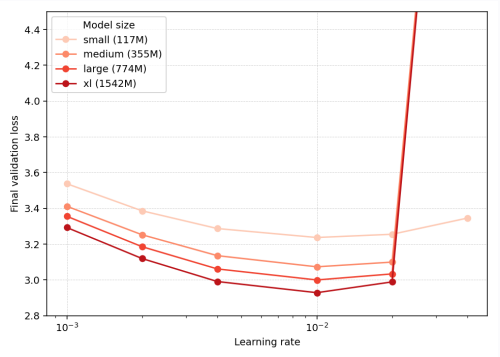
RowNorm

One row, one logit.
Untied embedding and readout.

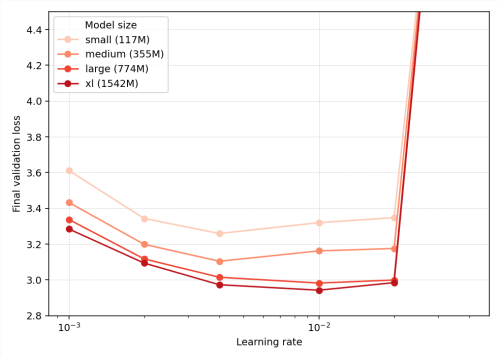
Control the selected embedding column and every output logit. Cross entropy uses the logit maximum.

Learning Rate Transfer Across Model Scales

Architecture: GPT-2 family (Small $\rightarrow$ XL)



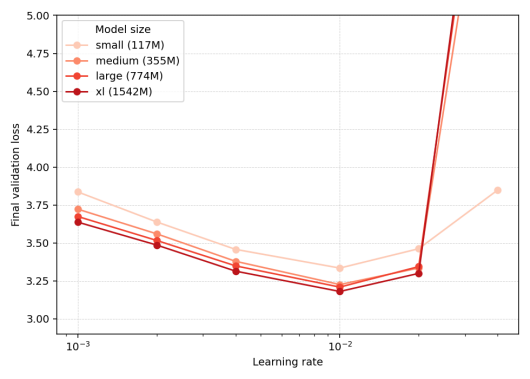
(a) MOGA ($p = 1.5$)



(b) MOGA ($p = 2$)

The optimal learning rate remains stable across model widths.

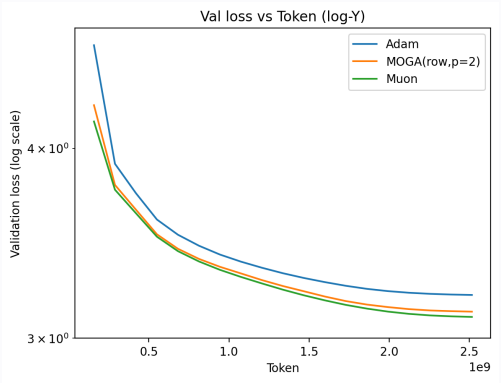
Learning Rate Transfer Across Model Scales



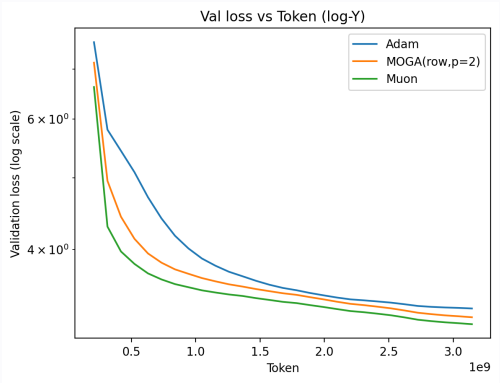
(c) MOGA ($p = 3$)

Learning rate transfer. The optimal learning rate remains stable from GPT-2 Small to XL.

LLM Pretraining: Standard Token Budget



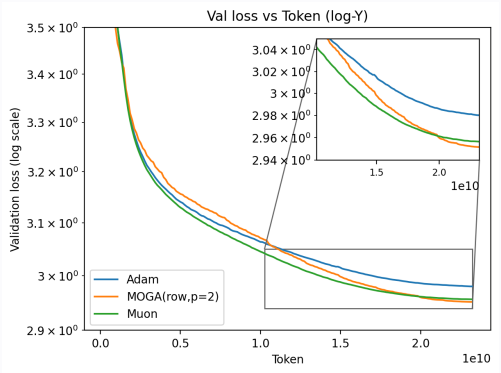
LLaMA 130M



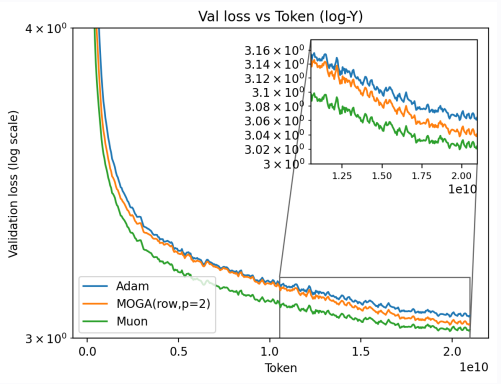
GPT-2 Small model

MOGA converges at nearly the same rate as Muon under a $1\times$ Chinchilla budget.

LLM Pretraining: Large Token Budget



LLaMA 130M



GPT-2 Small model

MOGA converges faster late in training, when the loss is small.

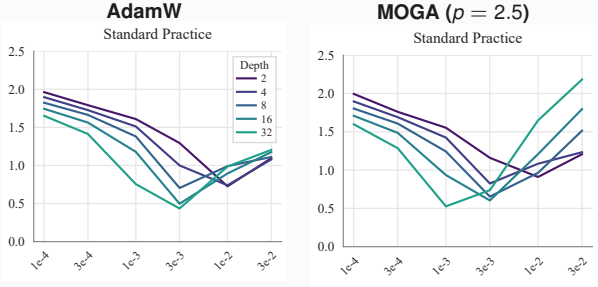
Related Work on Row and Column Normalization

- Training Deep Learning Models with Norm-Constrained LMOs, arXiv:2502.07529
- A Minimalist Optimizer Design for LLM Pretraining, arXiv:2506.16659
- Mano: Rethinking Manifold Optimization for LLM Training, arXiv:2601.23000
- RMNP: Row-Momentum Normalized Preconditioning for Scalable Matrix-Based Optimization, arXiv:2603.20527

Depth Transfer: Naive Scaling Can Fail

Residual MLPs on CIFAR10. Fixed width, two linear layers per residual branch.

$$y_k = y_{k-1} + \alpha f_{W_k}(y_{k-1}), \qquad \text{naive residual multiplier: } \alpha = 1.$$



Training loss versus learning rate, $K = 2$ to 32.

Updates propagate through later blocks

$$J_k = I + \alpha\, \partial_y f_{W_k},$$

$$\|J_K \cdots J_j\| \le (1 + \alpha L_f)^K.$$

With $\alpha = 1$, this worst case bound grows with depth.

The best tested learning rate drifts as depth increases.

Width scaling alone does not control accumulation over depth.

Source: *On the Depth Transfer of Neural Optimizers via L Smoothness*, Fig. 1 and Sec. 3.1.

A Smoothness Bound Independent of Depth

Fixed width. Bounded inputs and parameters. Uniform C^2 regularity of residual branches and loss.

$$\alpha = \frac{\beta}{K}, \quad \beta = \Theta(1), \qquad \|\Delta W\|_{\text{block}} = \max_k \|\Delta W_k\|_P.$$

Controlled propagation

$$\|J_K \cdots J_j\| \le \left(1 + \frac{\beta L_f}{K}\right)^K \le e^{\beta L_f}.$$

$$\|D_{W_j} y_K\| = O(K^{-1}), \qquad \|D_W y_K\| = O(1).$$

Each block contributes $O(1/K)$ to the total first variation.

All second order interactions

Derivatives of y_K	Count	Size per term
Same block	K	$O(K^{-1})$
Different blocks	$K(K-1)$	$O(K^{-2})$

$$K\,O(K^{-1}) + K(K-1)\,O(K^{-2}) = O(1).$$

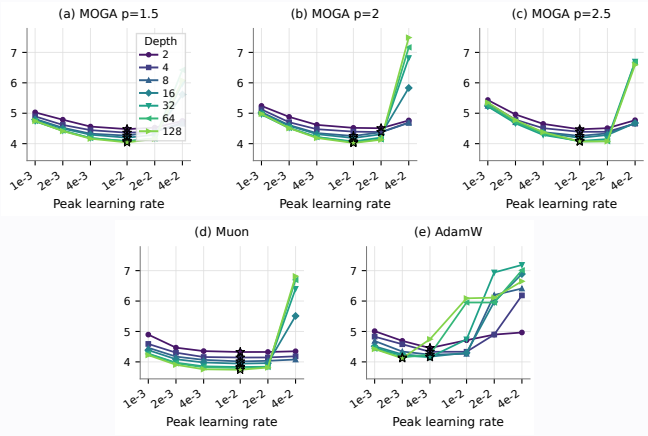
Theorem: depth independent L smoothness

$$\|\nabla \mathcal{L}(W) - \nabla \mathcal{L}(V)\|_{\text{block},*} \le L\|W - V\|_{\text{block}}, \qquad L = O(1) \text{ in } K.$$

Source: *On the Depth Transfer of Neural Optimizers via L Smoothness*, Theorems 1 and 2, Sec. 3.3.

Learning Rate Transfer Across Depth

GPT2, width 256, depths 2 to 128. Final training cross entropy on OpenWebText.



Muon and MOGA

Muon and MOGA ($p = 1.5, 2, 2.5$) keep their best tested rates close to 10^{-2} over a $64\times$ depth range.

AdamW

A common useful region remains. The best tested rate shifts from 4×10^{-3} to 2×10^{-3} at greater depths.

ResNet replication

MOGA shares the best tested training rate across 8, 16, and 32 residual blocks on CIFAR100.

Source: *On the Depth Transfer of Neural Optimizers via L Smoothness*, Figs. 2 and 3. Stars mark grid minima.

Take Home Messages

- **Hyperparameter transfer** is essential for optimizer design at LLM scale.
- **Matrix operator geometry** unifies SignSGD, AdamW, Muon, and normalized matrix updates.
- **Mean normalized geometry** yields width independent Lipschitz and smoothness bounds when $q \geq 2p$, together with width aware learning rate scaling.
- **Depth transfer:** residual scaling $\alpha = \beta/K$ controls curvature across depth in all prescribed optimizer geometries at fixed width.

Future Direction: Is there a norm unifies approxiamtion, generalization and optimization?

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Appendix: Spectral Condition

- **Feature scale**

$$\|y_l\|_2 = \Theta(\sqrt{\text{fanout}}), \text{ and } \|\Delta y_l\|_2 = \Theta(\sqrt{\text{fanout}})$$

- **Parameter scale**

$$\|W_l\|_2 = \Theta\left(\sqrt{\frac{\text{fanout}}{\text{fanin}}}\right), \text{ and } \|\Delta W_l\|_2 = \Theta\left(\sqrt{\frac{\text{fanout}}{\text{fanin}}}\right)$$

Yang G, Simon J B, Bernstein J. A spectral condition for feature learning. arXiv preprint arXiv:2310.17813, 2023.

The Operation Determines the Second Derivative

Nonlinear modules contribute Hessian terms. Their structure determines the needed norms.

Coordinatewise activation

$$D^2\sigma(z)[u,v] = \sigma''(z) \odot u \odot v.$$

For $u = v = \sqrt{m}\, e_1$, both RMS norms are 1, but

$$\|u \odot v\|_{\mathrm{RMS}} = \sqrt{m}.$$

For SiLU at zero,

$$\|D^2\sigma(0)[u,v]\|_{\mathrm{RMS}} = \sqrt{m}/2.$$

The earlier $q \geq 2p$ condition controls this coordinate product.

Normalized inner product

$$B(q,k) = \frac{q^\top k}{r}, \qquad D_q D_k B[u,v] = \frac{u^\top v}{r}.$$

Cauchy–Schwarz gives

$$\left|\frac{u^\top v}{r}\right| \leq \|u\|_{\mathrm{RMS}} \|v\|_{\mathrm{RMS}}.$$

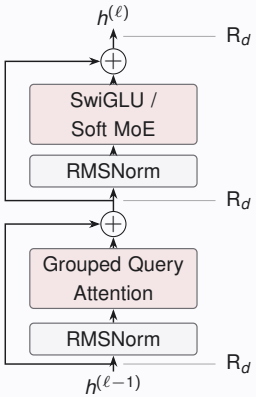
For $u = v = \sqrt{r}\, e_1$, the mixed derivative equals 1.

A normalized contraction can preserve RMS geometry.

The distinction used below

Gate retains coordinatewise products. Attention contracts coordinates into scores and applies softmax in a maximum norm.

A Concrete Decoder with Compatible Feature Spaces



Feature coordinates

$$\mathrm{R}_n: \quad \|x\|_{\mathrm{RMS}} = \|x\|_2/\sqrt{n}, \qquad \mathrm{C}_n: \quad \|x\|_\infty.$$

Two different norms across heads

$$\|q\|_{\mathsf{A}_{H,r}} = \max_h \|q_h\|_{\mathrm{RMS}}, \qquad \text{Q/K},$$

$$\|v\|_{\mathsf{H}_{H,r}} = \left(\frac{1}{H}\sum_h \|v_h\|_{\mathrm{RMS}}^2\right)^{1/2}, \qquad \text{V/O}.$$

All sequence features additionally use a maximum over tokens.

Architecture and domain for this certificate

Fixed depth and bounded parameter norms. RMSNorm has fixed $\varepsilon > 0$ and bounded coordinate gains. Q/K use RMSNorm and $1/r$ scores. Heads share the allowed key set and GQA replication is balanced. Embedding and readout are separate parameters.

RMSNorm Has a Dimension Independent Hessian

Write $s = (\varepsilon + \|x\|_{\mathrm{RMS}}^2)^{1/2}$, $n(x) = x/s$, and $\langle u, v\rangle_{\mathrm{RMS}} = u^\top v/n$ on $\mathbb{R}^n$.

$$D^2 n(x)[u,v] = \frac{-1}{s^3}(u\langle x,v\rangle_{\mathrm{RMS}} + v\langle x,u\rangle_{\mathrm{RMS}} + x\langle u,v\rangle_{\mathrm{RMS}}) + \frac{3x}{s^5}\langle x,u\rangle_{\mathrm{RMS}}\langle x,v\rangle_{\mathrm{RMS}}.$$

Every quadratic factor is a normalized inner product. Since $\|x\|_{\mathrm{RMS}} \le s$,

$$\boxed{\|D^2 n(x)[u,v]\|_{\mathrm{RMS}} \le \frac{6}{\varepsilon}\|u\|_{\mathrm{RMS}}\|v\|_{\mathrm{RMS}}.}$$

For the gain $N(x,\gamma) = \gamma \odot n(x)$, with perturbations (u,a) and (v,b),

$$D^2 N = \gamma \odot D^2 n[u,v] + a \odot Dn[v] + b \odot Dn[u].$$

The gain and its perturbations use C_n: one RMS factor times bounded coordinate multipliers.

Other operations at the same site

Fixed orthogonal RoPE preserves RMS. It and residual addition have zero Hessian.

Q/K Need RMS Control within Each Head

Here q, k include QK RMSNorm and fixed RoPE. Subscripts a, b, ab denote parameter variations.

Scores and their variations

$$z = \frac{q^\top k}{r},$$

$$z_a = \frac{q_a^\top k + q^\top k_a}{r},$$

$$z_{ab} = \frac{q_{ab}^\top k + q_a^\top k_b}{r} + \frac{q_b^\top k_a + q^\top k_{ab}}{r}.$$

Every term is a normalized inner product:

$$|u^\top v|/r \le \|u\|_{\mathrm{RMS}} \|v\|_{\mathrm{RMS}}.$$

The RMSNorm bounds therefore control z, z_a, and z_{ab} in maximum norm over keys and heads.

Why the rule is Muon

$$\|Q\|_{\mathsf{A}_{H,r}} = \max_h \|q_h\|_{\mathrm{RMS}}.$$

For stored blocks $W_h \in \mathbb{R}^{r\times d}$,

$$N_Q(W_Q) = \sqrt{d/r}\, \max_h \|W_h\|_{\mathrm{op}}.$$

The ball separates across heads. Its support direction is

$$\boxed{D_h = \sqrt{r/d}\ \mathrm{polar}(G_h).}$$

Each head has its own spectral norm constraint. RowNorm would instead constrain every output coordinate.

The head maximum also controls perturbations after normalization. With $1/\sqrt{r}$ scores, this estimate acquires a $\sqrt{r}$ factor when r grows.

Softmax Acts on Scores in Maximum Norm

Let $\pi = S(z) = \mathrm{softmax}(z)$ and $\bar{a} = \sum_j \pi_j a_j$. Direct differentiation gives

$$DS(z)[a]_j = \pi_j(a_j - \bar{a}),$$
$$D^2S(z)[a,b]_j = \pi_j\big((a_j - \bar{a})(b_j - \bar{b}) - \mathrm{Cov}_\pi(a,b)\big).$$

Cauchy–Schwarz under the probability law π yields

$$\|DS(z)[a]\|_1 \le \|a\|_\infty, \qquad \boxed{\|D^2S(z)[a,b]\|_1 \le 2\|a\|_\infty\|b\|_\infty.}$$

These constants do not depend on the number of keys. The probability norm is ordinary ℓ^1.

The full chain rule still contains both second order terms

$$D^2(S \circ B)[u,v] = D^2S(B)[DB[u], DB[v]] + DS(B)[D^2B[u,v]].$$

Where the stronger control enters

The score map $B(q,k) = q^\top k/r$ supplies maximum norm bounds from head RMS bounds. Softmax uses those bounds to produce controlled probability variations in ℓ^1.

A Common Key Distribution Preserves Value RMS

For one query i, write $o_h = \sum_j \omega_{hj} v_{jh}$ and use head RMS $\|\cdot\|_{\mathcal{H}}$. Bounded scores $|z_{hj}| \le B$ give a distribution shared by every head:

$$\omega_{hj} \le D\mu(j), \qquad D = e^{2B}, \qquad \mu(j) = \frac{1}{|J_i|}.$$

Triangle inequality within each head and Minkowski across heads give

$$\boxed{\|o\|_{\mathcal{H}} \le D \sum_j \mu(j) \|v_j\|_{\mathcal{H}} \le D \max_j \|v_j\|_{\mathcal{H}}.}$$

The same proof controls the Hessian:

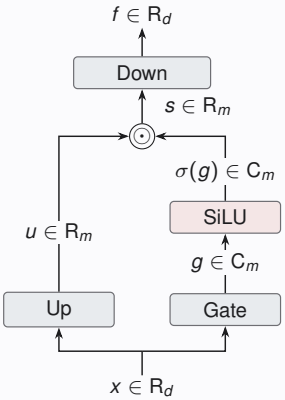
$$o_{ab} = \sum_j \left(\omega_j v_{j,ab} + \omega_{j,a} v_{j,b} + \omega_{j,b} v_{j,a} + \omega_{j,ab} v_j\right).$$

The preceding score bounds imply $|\omega_{j,a}| \le C_1 \mu(j)$ and $|\omega_{j,ab}| \le C_2 \mu(j)$, uniformly over heads.

Why V and O use whole matrix Muon

Head RMS equals flattened RMS and is preserved by balanced GQA replication. Thus V and O connect RMS spaces. The common key bound controls concentration across heads.

SwiGLU Keeps the Stronger Norm on the Gate



For $s = u \odot \sigma(g)$, $\sigma = \mathrm{SiLU}$,

$$\begin{aligned} s_{ab} = {} & u_{ab} \odot \sigma(g) \\ & + \sigma'(g) \odot (u_a \odot g_b + u_b \odot g_a) \\ & + u \odot \sigma'(g) \odot g_{ab} \\ & + u \odot \sigma''(g) \odot g_a \odot g_b. \end{aligned}$$

Every term contains one RMS factor. Gate factors use maximum norm:

$$\|u \odot a \odot b\|_{\mathrm{RMS}} \le \|u\|_{\mathrm{RMS}} \|a\|_\infty \|b\|_\infty.$$

Up and Down: RMS to RMS, whole matrix Muon.

Gate: RMS to coordinate maximum, RowNorm.

At $g = 0$, $u = \mathbf{1}$, and $g_a = g_b = \sqrt{m}\, e_1$, the red term has RMS $\sqrt{m}/2$. The original curvature obstruction remains on this branch.

Routing and Loss Use Probability Geometry

Soft mixture of experts

$$s = W_R x \in \mathrm{C}_N, \qquad \pi = \mathrm{softmax}(s).$$

For $y = \sum_e \pi_e F_e$, router RowNorm controls score variations in maximum norm. Softmax gives probability variations in ℓ^1.

$$y_{ab} = \sum_e \big(\pi_e F_{e,ab} + \pi_{e,a} F_{e,b} + \pi_{e,b} F_{e,a} + \pi_{e,ab} F_e\big).$$

Every term uses

$$\Big\| \sum_e a_e f_e \Big\|_{\mathrm{RMS}} \le \|a\|_1 \max_e \|f_e\|_{\mathrm{RMS}}.$$

There is no factor of the expert count. This uses a full softmax router.

Readout and cross entropy

$$z = W_{\mathrm{lm}} x \in \mathrm{C}_\nu, \qquad \ell(z) = \log \sum_j e^{z_j} - z_y.$$

Readout RowNorm supplies the logit maximum norm. With $p = \mathrm{softmax}(z)$,

$$\|\nabla \ell(z)\|_1 \le 2, \qquad D^2 \ell(z)[a,b] = \mathrm{Cov}_p(a,b).$$

Hence

$$|D^2 \ell(z)[a,b]| \le \|a\|_\infty \|b\|_\infty.$$

Along a parameter perturbation,

$$|(\ell \circ z)_{ab}| \le \|z_a\|_\infty \|z_b\|_\infty + 2\|z_{ab}\|_\infty.$$

Vocabulary size does not enter this estimate.

Embedding is linear, with a column norm from ordinary vocabulary ℓ^1 to residual RMS. Averaging the loss over tokens preserves the bounds.

The Feature Spaces Determine the Matrix Updates

The update subtracts the support direction, which maximizes $\langle G, D \rangle$ over the unit ball.

Parameter	Feature interface	Positive support direction
Q/K	$\mathrm{R}_d \to \mathrm{A}_{H,r}$, head maximum	$\sqrt{r/d}\ \mathrm{polar}(G_h)$, each head
V	$\mathrm{R}_d \to \mathrm{H}_{H_{kv},r}$	$\sqrt{H_{kv}r/d}\ \mathrm{polar}(G_V)$
O	$\mathrm{H}_{H_q,r} \to \mathrm{R}_d$	$\sqrt{d/(H_q r)}\ \mathrm{polar}(G_O)$
Up / Down	$\mathrm{R}_d \to \mathrm{R}_m$ / $\mathrm{R}_m \to \mathrm{R}_d$	$\sqrt{m/d}\ \mathrm{polar}(G_{\mathrm{up}})$ / $\sqrt{d/m}\ \mathrm{polar}(G_{\mathrm{down}})$
Gate / router / LM	$\mathrm{R}_d \to \mathrm{C}_{\mathrm{out}}$	$d^{-1/2}\ \mathrm{row}(G)$
Embedding / gain	$\ell^1 \to \mathrm{R}_d$ / coordinate gain	$\sqrt{d}\ \mathrm{col}(G_E)$ / $\mathrm{sign}(G_\gamma)$

A module by module curvature certificate

Under the stated assumptions, values and their first two variations stay bounded through every module. Fixed depth composition gives width independent curvature.

Polar is the ideal direction underlying Muon. The following experiments evaluate the original MOGA variants.

$\|\cdot\|_{p\to q}$

Thank you

Questions?

Yiping Lu

Beijing International Center for Mathematical Research
Peking University